



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Harman Singh
10 September 2026



Outline

- Executive Summary
- Introduction & business problem
- Methodology: API, scraping, wrangling, EDA, SQL, Folium, Dash, classification
- Results: visual EDA, SQL, interactive analytics and model performance
- Conclusions, innovative insights and appendix

Executive Summary

Falcon 9 records

90

Analyzed launch records

Overall landing success

66.7%

90 launches; Class = 1

Top launch volume

55

CCAFS SLC 40

Best test accuracy

83.33%

LR / SVM / KNN tie

- Combined SpaceX REST API data with Wikipedia launch records, then cleaned and transformed the data for analysis.
- EDA and SQL revealed strong time, site, orbit and payload patterns in first-stage landing outcomes.
- Interactive Folium and Plotly Dash views translated the findings into geographic and operational decision support.
- Four classifiers were evaluated; the standard capstone test split produced 83.33% accuracy for Logistic Regression, SVM, Decision Tree and KNN.

Introduction

Project background & context

- SpaceX reduced launch costs through Falcon 9 first-stage reusability.
- The project treats launch history as a data science problem: understand the drivers of landing success and predict future outcomes.
- Target outcome: a reusable, evidence-based framework for evaluating first-stage landing risk before launch.

Questions addressed

- Which launch sites and orbits are associated with higher landing success?
- How do payload mass and flight number relate to landing outcome?
- Can a classification model predict whether the first stage will land successfully?

Section 1

Methodology

Methodology

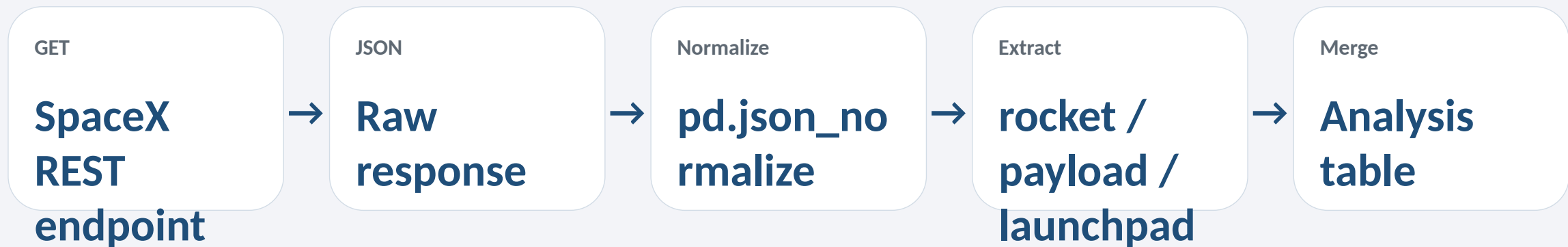


End-to-end workflow: raw launch records → analysis-ready landing labels → descriptive insights → interactive decision tools → predictive classification.

Data Collection



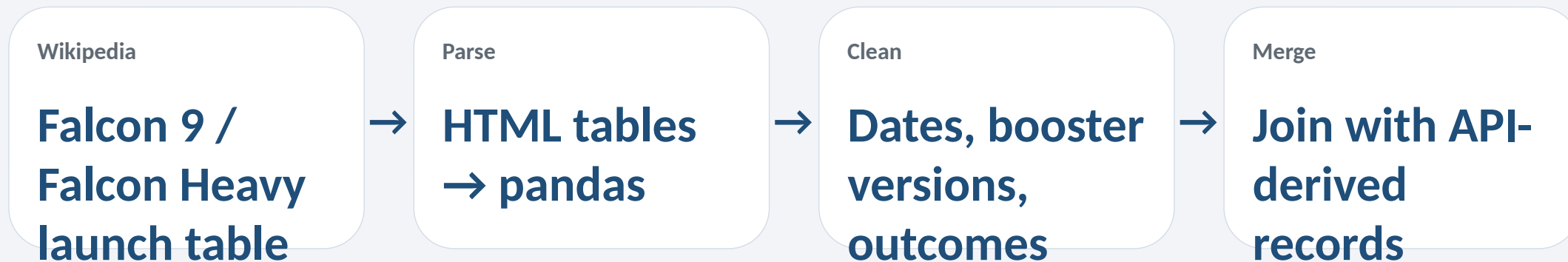
Data Collection – SpaceX API



- Static-fire and launch timestamps were retained for temporal analysis.
- Falcon 1 records were removed so the modeling target represented Falcon 9 first-stage landings.
- Landing-pad, payload and launch-site fields were preserved for downstream EDA.

Notebook reference **/ core**
GitHub URL: [YOUR_REPOSITORY_URL](#)

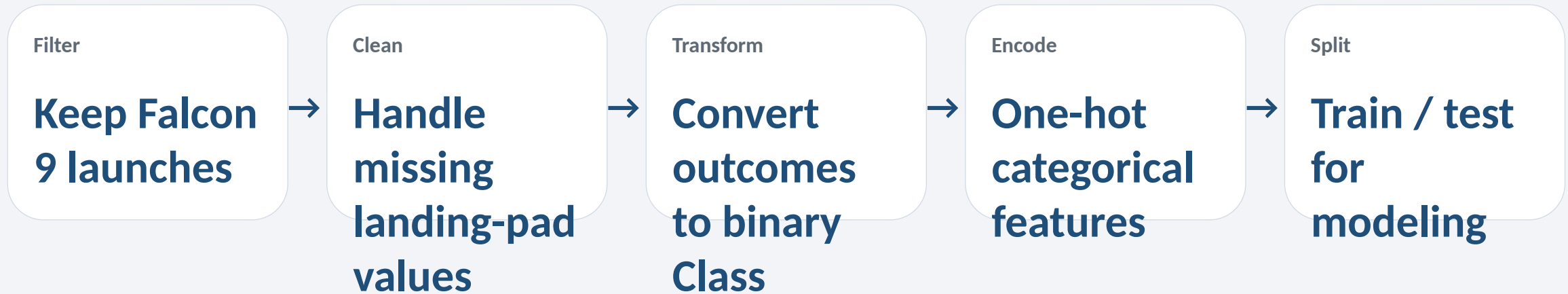
Data Collection - Scraping



Web scraping supplied mission-level fields that complement the API, especially launch / booster / outcome information used to define the supervised learning label.

GitHub URL: [YOUR_REPOSITORY_URL](#)

Data Wrangling



Target label: Class = 1 for successful first-stage landing; Class = 0 for unsuccessful / no landing. Numeric fields were retained for payload and flight-history effects.

GitHub URL: [YOUR_REPOSITORY_URL](#)

EDA with Data Visualization

- Scatter plots: Flight Number vs. Launch Site; Payload vs. Launch Site; Flight Number vs. Orbit; Payload vs. Orbit.
- Bar chart: success rate by orbit type.
- Line chart: yearly average landing success.
- Color / marker encoding separates successful and unsuccessful landings.
- Why these charts? They expose temporal trends, geographic differences, mission-profile effects and potential non-linear relationships.
- Visual EDA guided feature selection and the predictive modeling stage.
- Notebook reference: `YOUR_REPOSITORY_URL`

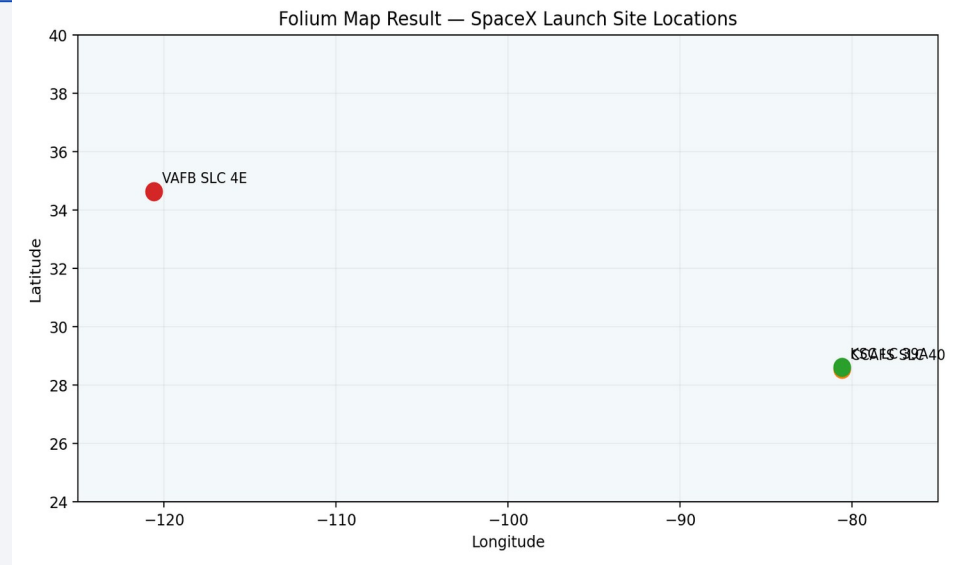
EDA with SQL

- SELECT DISTINCT → enumerate unique launch sites.
- LIKE 'CCA%' + LIMIT → inspect CCAFS launch records.
- SUM / AVG → quantify payload capacity and average F9 v1.1 payload.
- MIN → locate the first successful ground-pad landing.
- WHERE + payload range → identify successful drone-ship landings between 4,000–6,000 kg.
- GROUP BY / COUNT → summarize mission outcomes and landing outcomes over time.
- Subquery + MAX → identify boosters carrying the maximum payload.

GitHub URL: [YOUR_REPOSITORY_URL](#)

Build an Interactive Map with Folium

- Marker objects identify launch-site coordinates.
- MarkerCluster groups nearby launch outcomes for readability.
- Green / red outcome markers distinguish successful vs. unsuccessful landings.
- Proximity analysis uses circles / lines and distance annotations to connect launch sites with coastlines, highways, railways and nearby cities.



Purpose: translate latitude/longitude into an operational view of launch-site geography and logistics.

Build a Dashboard with Plotly Dash

- Dropdown: select a launch site or view all sites.
- Pie chart: success / failure distribution.
- Payload scatter: compare payload mass against landing outcome.
- Range slider: focus on different payload bands.
- Dashboard goal: let stakeholders explore the same evidence without rerunning notebooks.

Interactive design

Inputs

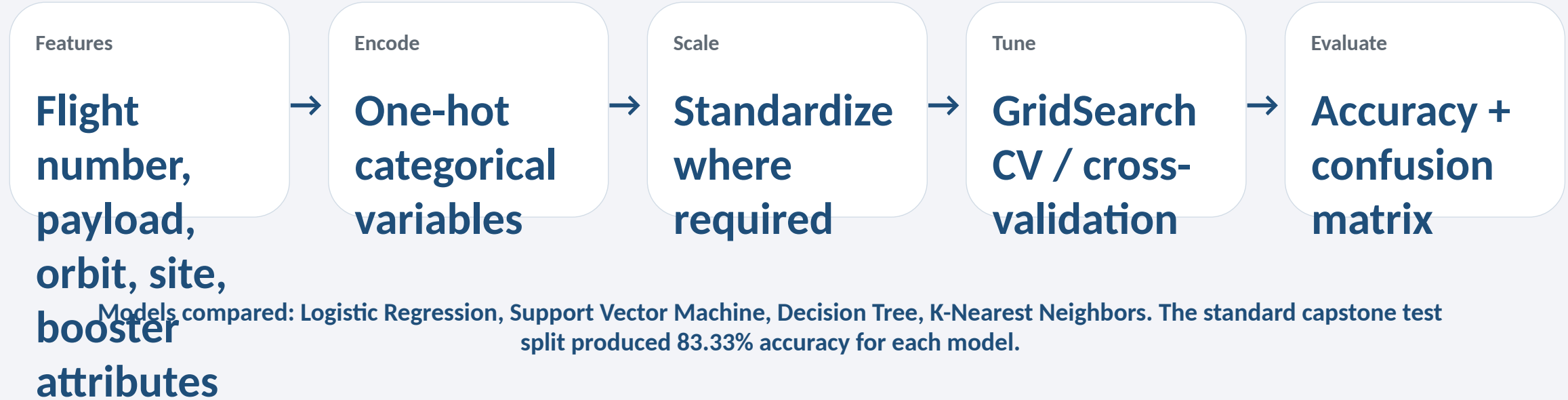
**Launch site + payload
range**

Outputs

Pie + scatter + success ratio

GitHub URL: YOUR_REPOSITORY_URL

Predictive Analysis (Classification)



GitHub URL: [YOUR_REPOSITORY_URL](#)

Results

EDA

**Launch, payload,
orbit & time
patterns**

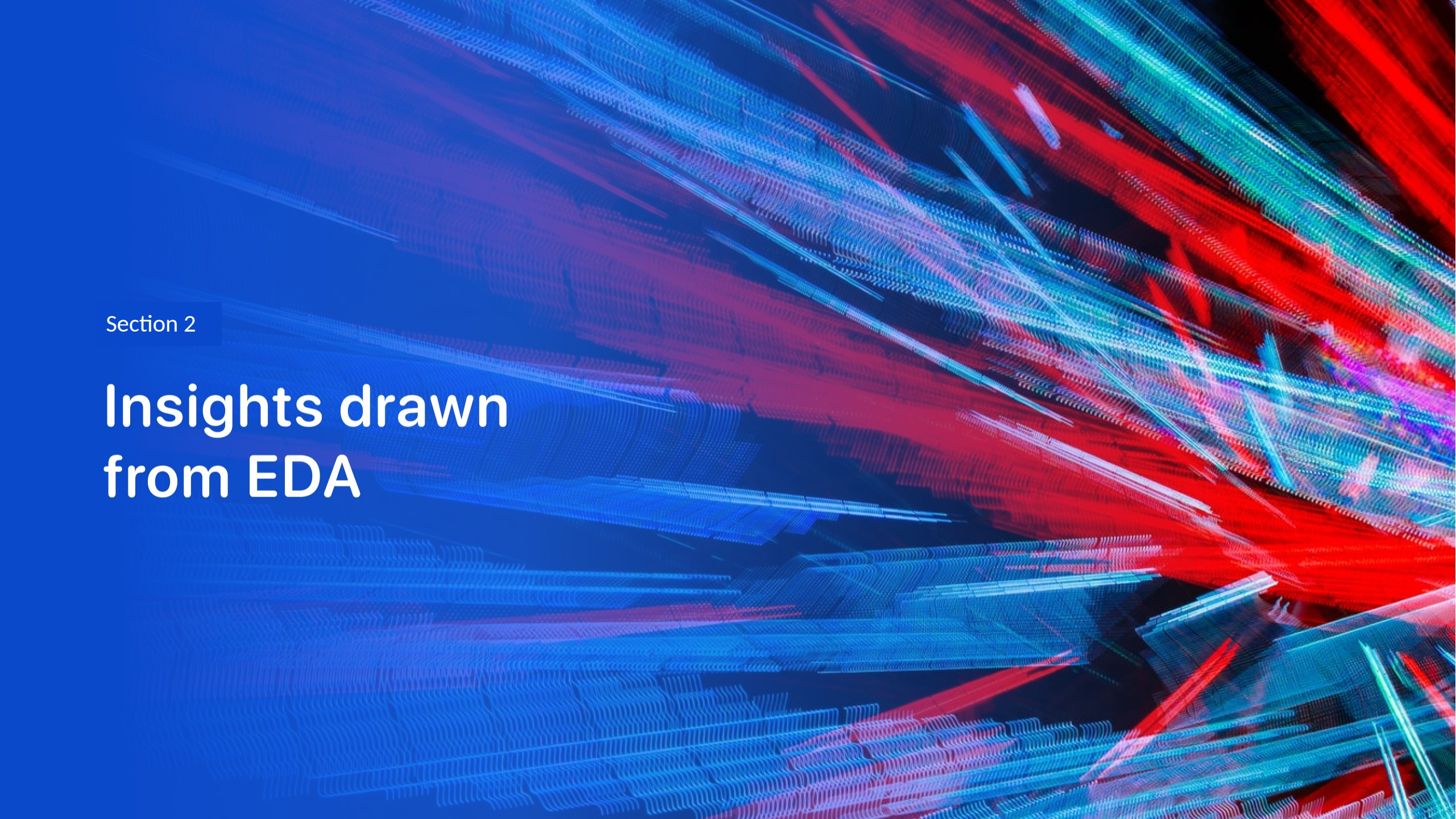
Interactive analytics

**Folium + Plotly
Dash**

Predictive analysis

**83.33% test
accuracy**

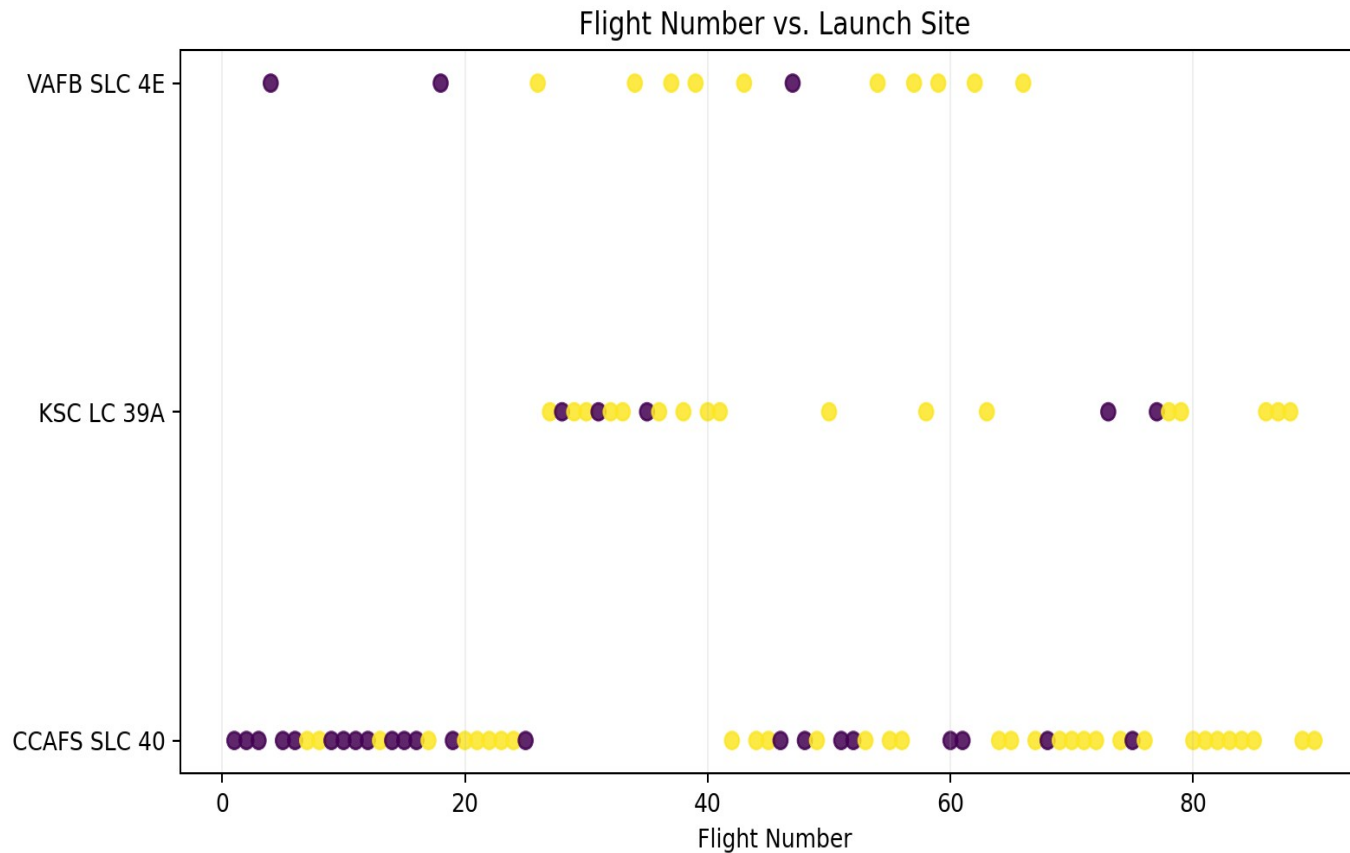
The results section moves from descriptive evidence to interactive exploration and finally to a predictive model for first-stage landing success.



Section 2

Insights drawn from EDA

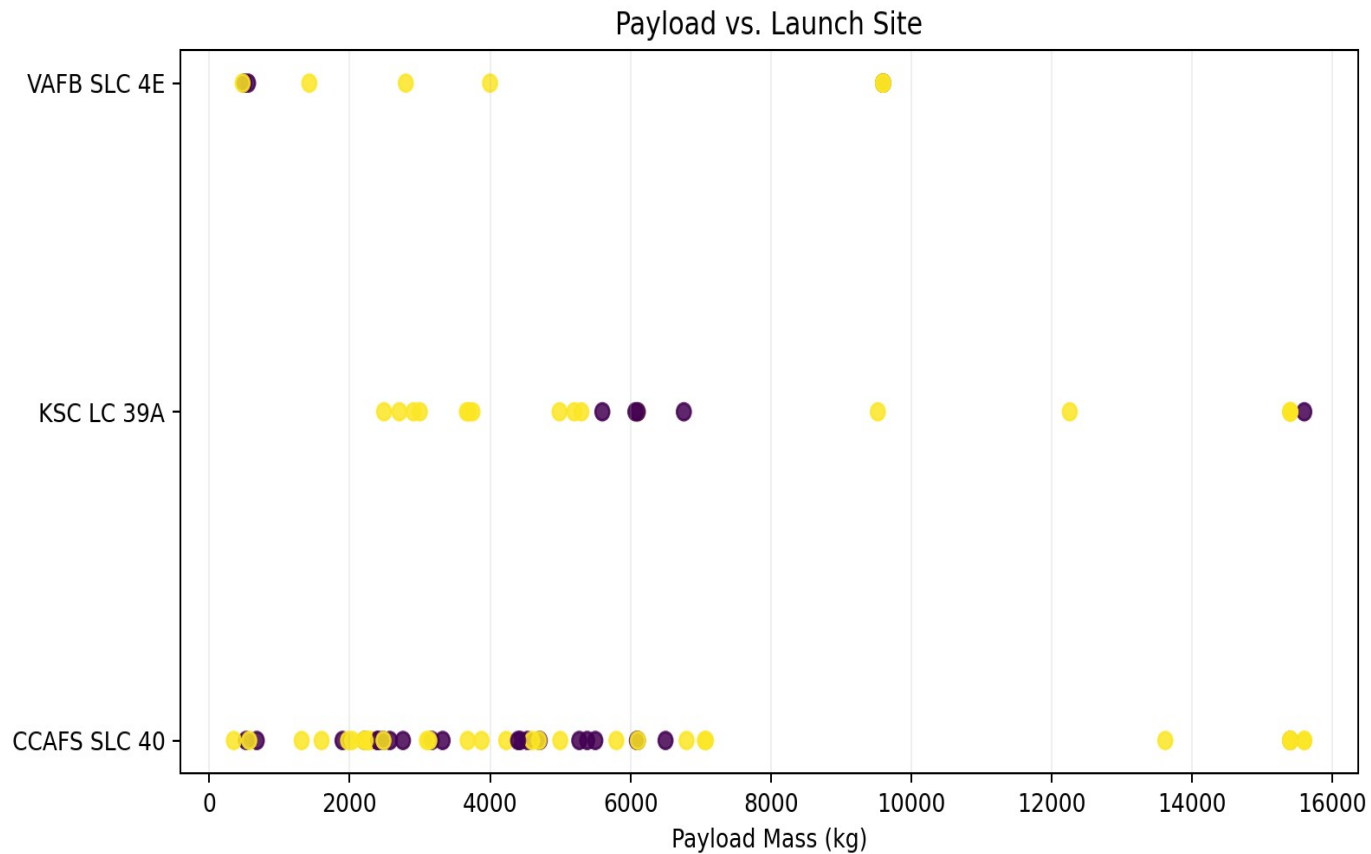
Flight Number vs. Launch Site



Key findings

- Early flights were concentrated at CCAFS SLC 40 and show more failures.
- After roughly flight 30, successful landings become much more frequent.
- KSC LC-39A appears later in the timeline and has a high success share.

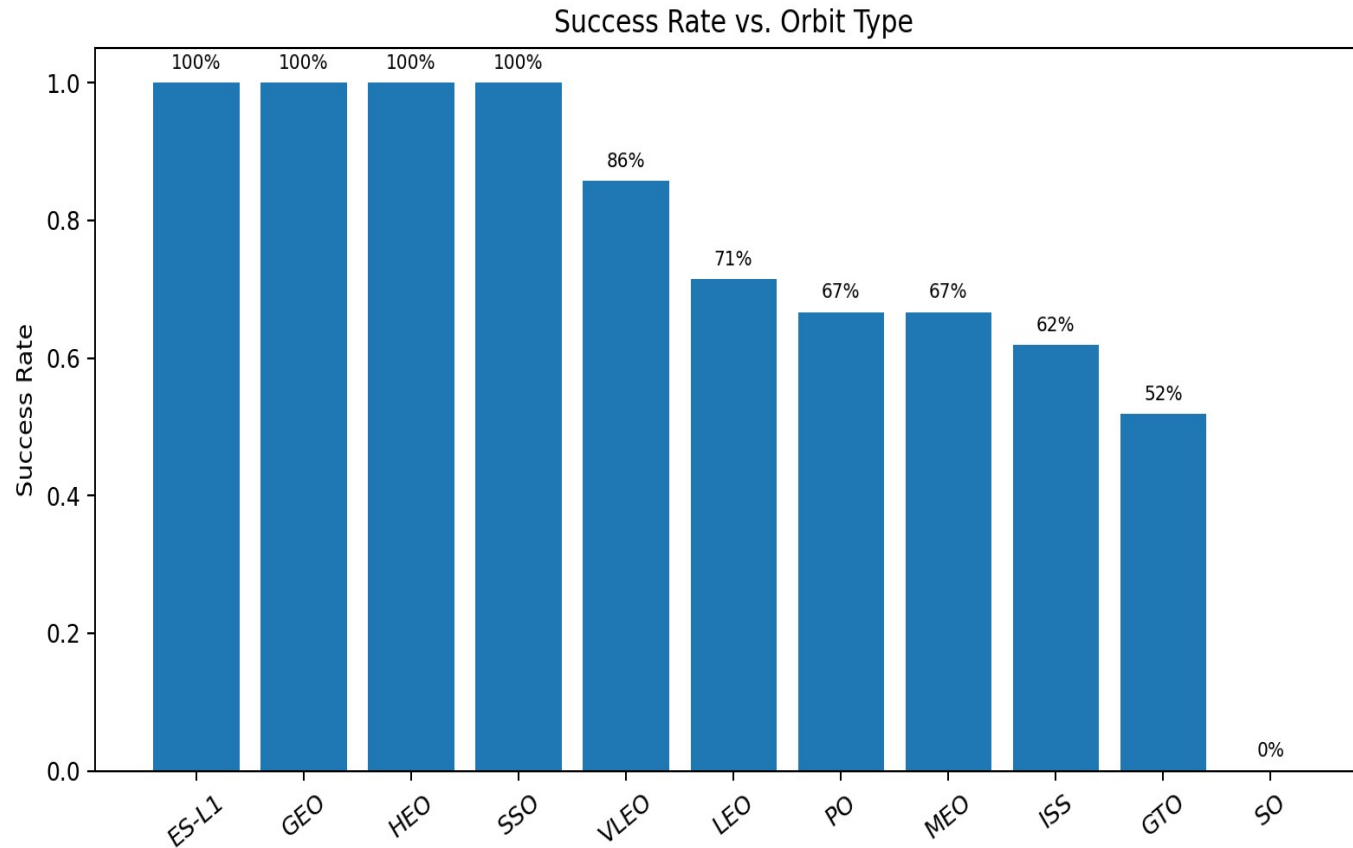
Payload vs. Launch Site



Key findings

- Most launches carry relatively light-to-medium payloads.
- Payloads above ~7,000 kg show few failures in this sample, but the number of observations is smaller.
- Launch site and payload distributions are not identical, supporting site-aware modeling.

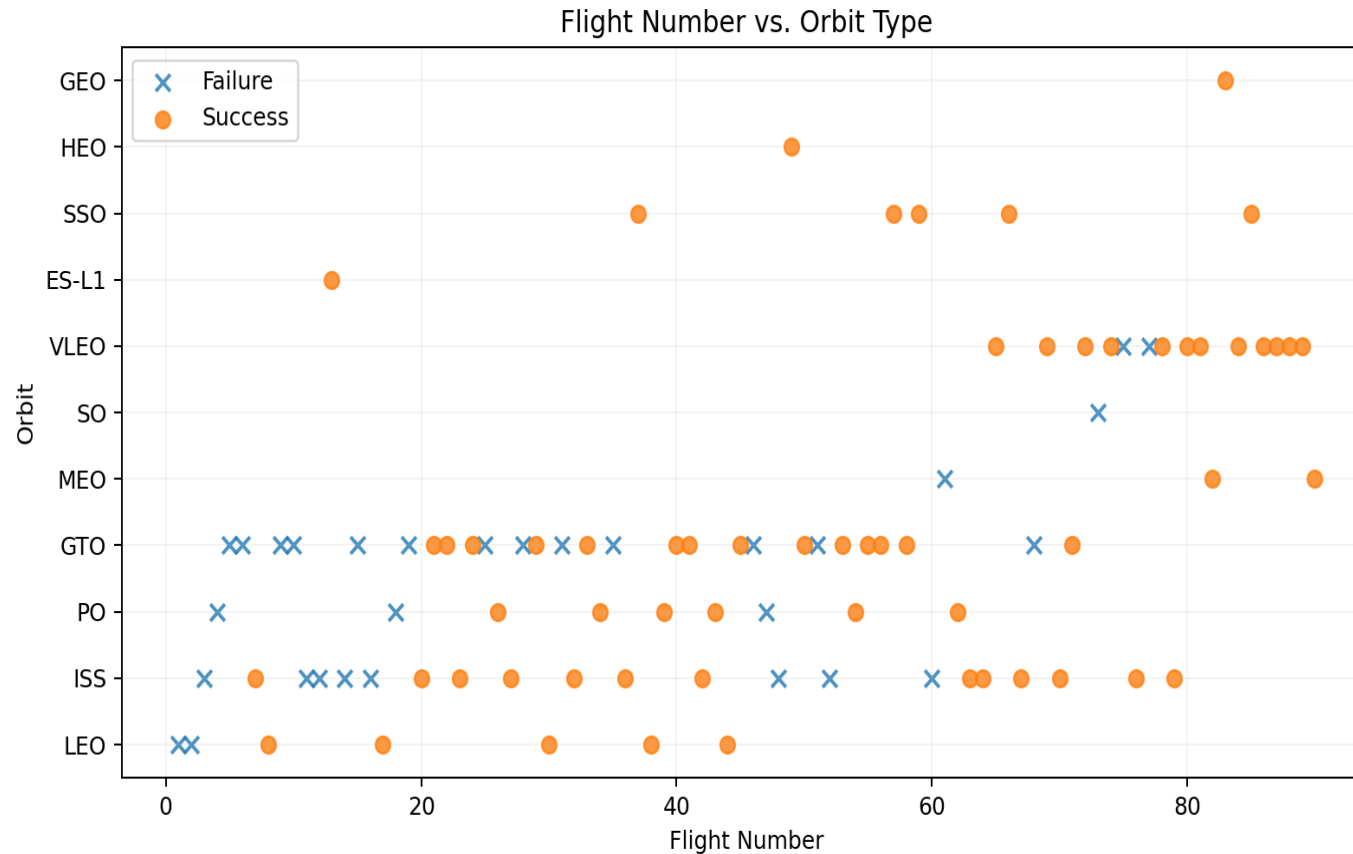
Success Rate vs. Orbit Type



Key findings

- ES-L1, GEO, HEO and SSO show 100% success in this sample.
- SO shows 0% success; GTO is the lowest among the frequently used orbital classes at ~52%.
- Rare-orbit rates should be interpreted cautiously because some have very few observations.

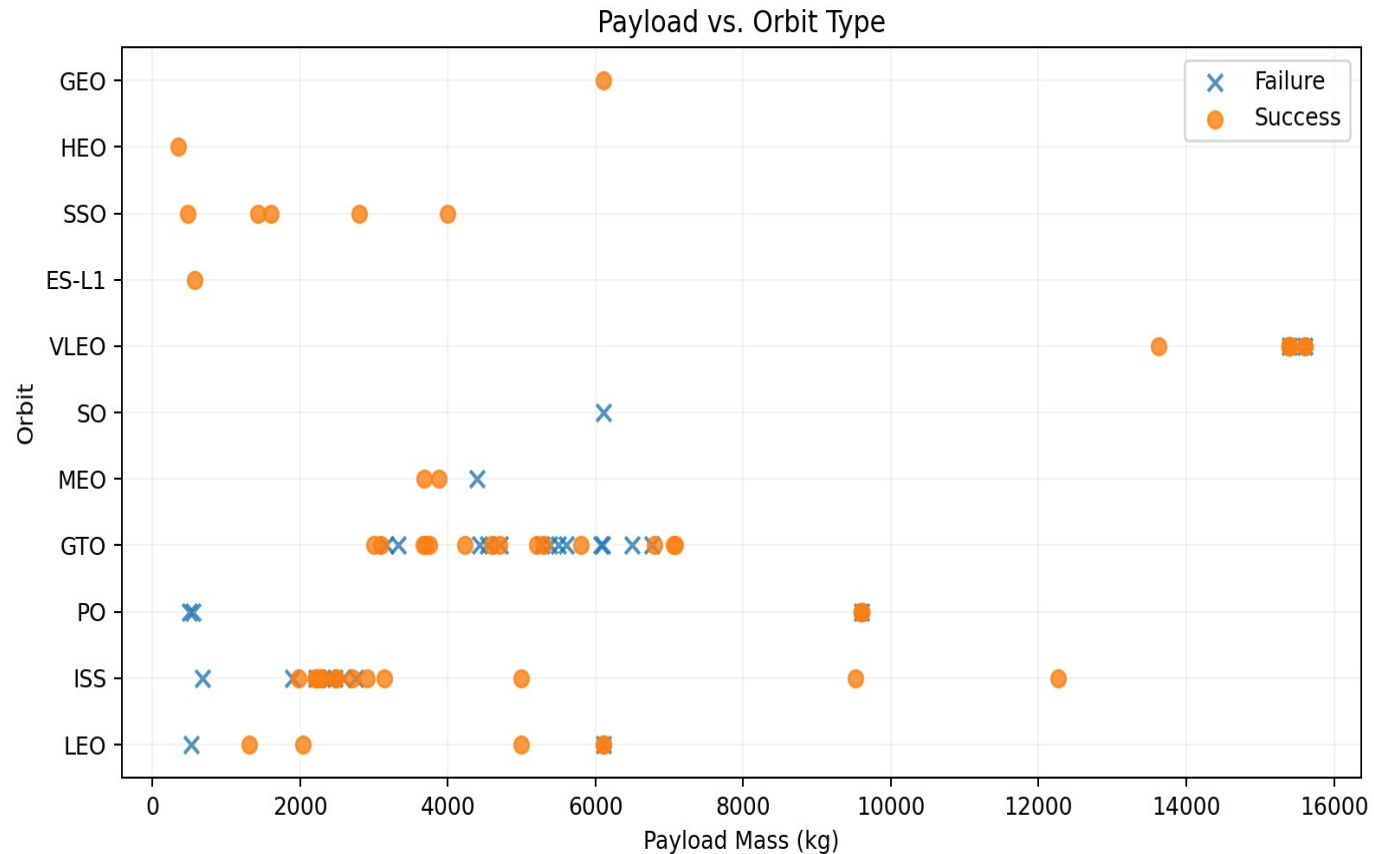
Flight Number vs. Orbit Type



Key findings

- GEO, HEO and ES-L1 have only one successful flight each, explaining their 100% rate.
- LEO shows a clearer time-related improvement in landing outcomes.
- GTO remains mixed across the flight timeline.

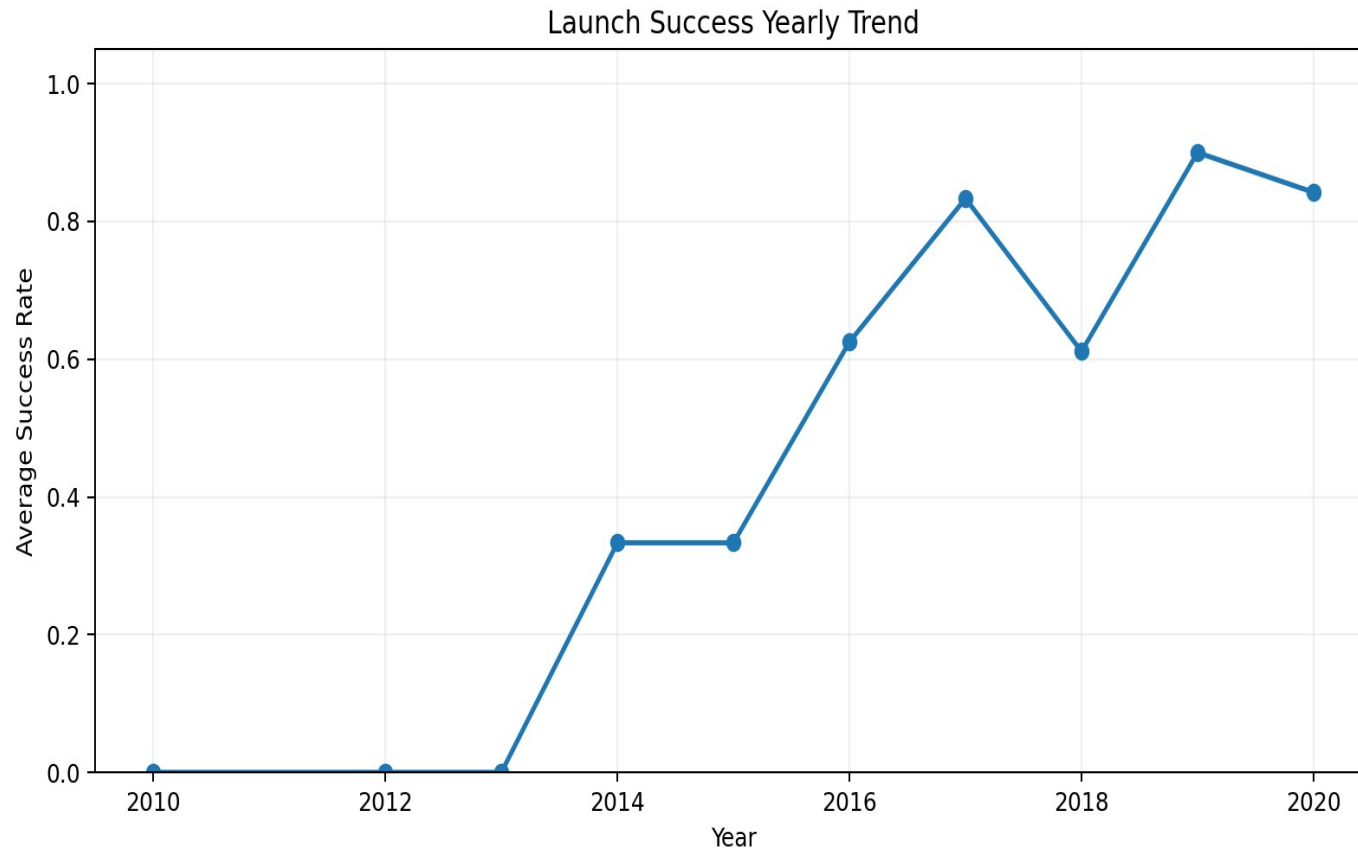
Payload vs. Orbit Type



Key findings

- Most payloads lie below 6,000 kg, with a second concentration at very high VLEO payloads.
- Successful outcomes dominate the higher-payload VLEO region in later years.
- Orbit choice and payload should be considered jointly rather than independently.

Launch Success Yearly Trend



Key findings

- Landing success improved materially after 2013.
- The trend peaks around 2018 before remaining high in 2019.
- This temporal pattern suggests operational learning and technology maturation.

All Launch Site Names

Unique Launch Sites

Launch Site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

SQL result & interpretation

DISTINCT returns four unique launch-site labels in the SQL dataset. The two CCAFS naming variants are retained because they occur as separate text values.

Notebook: [YOUR_REPOSITORY_URL](#)

Launch Site Names Begin with 'CCA'

First 5 rows matching 'CCA%'

Flight	Date	Launch Site
1	2010-06-04	CCAFS SLC 40
2	2012-05-22	CCAFS SLC 40
3	2013-03-01	CCAFS SLC 40
5	2013-12-03	CCAFS SLC 40
6	2014-01-06	CCAFS SLC 40

SQL result & interpretation

LIKE 'CCA%' isolates records whose launch-site text begins with CCA; LIMIT 5 returns the first five matching records.

Notebook: [YOUR_REPOSITORY_URL](#)

Total Payload Mass

NASA (CRS) Total Payload

Total Payload
45,596 kg

SQL result & interpretation

NASA (CRS) missions carried 45,596 kg of payload in the analyzed SQL dataset.

Notebook: [YOUR_REPOSITORY_URL](#)

Average Payload Mass by F9 v1.1

Average Payload - F9 v1.1

Average Payload
2,534.67 kg

SQL result & interpretation

AVG(payload_mass__kg_) gives an average of approximately 2,534.67 kg for F9 v1.1 boosters.

Notebook: [YOUR_REPOSITORY_URL](#)

First Successful Ground Landing Date

First Successful Ground Landing

Date
2015-12-22

SQL result & interpretation

MIN(Date) identifies 2015-12-22 as the first successful ground-pad landing.

Notebook: [YOUR_REPOSITORY_URL](#)

Successful Drone Ship Landing with Payload between 4000 and 6000

Successful Drone-Ship Landings: 4-6 t

Booster	Payload (kg)
F9 FT B1022	4,696
F9 FT B1026	4,600
F9 FT B1021.2	5,300
F9 FT B1031.2	5,200

SQL result & interpretation

Four booster versions meet the successful drone-ship and payload-range criteria.

Notebook: [YOUR_REPOSITORY_URL](#)

Total Number of Successful and Failure Mission Outcomes

Mission Outcomes

Mission Status	Count
Success	100
Failure	1

SQL result & interpretation

The mission-outcome query reports 100 successful outcomes and 1 failure in the SQL dataset.

Notebook: [YOUR_REPOSITORY_URL](#)

Boosters Carried Maximum Payload

Boosters Carrying Maximum Payload

Booster Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

SQL result & interpretation

A MAX subquery identifies 12 booster versions associated with the maximum payload mass.

Notebook: [YOUR_REPOSITORY_URL](#)

2015 Launch Records

2015 Failed Drone-Ship Landings

Booster	Launch Site	Landing Outcome
F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

SQL result & interpretation

Two failed drone-ship landing attempts in 2015 occurred from CCAFS LC-40: B1012 and B1015.

Notebook: [YOUR_REPOSITORY_URL](#)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing Outcomes Ranked: 2010-06-04 to 2017-03-20

Landing Outcome	Count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

SQL result & interpretation

No attempt is the most frequent outcome in this early period; drone-ship success and failure each occur five times.

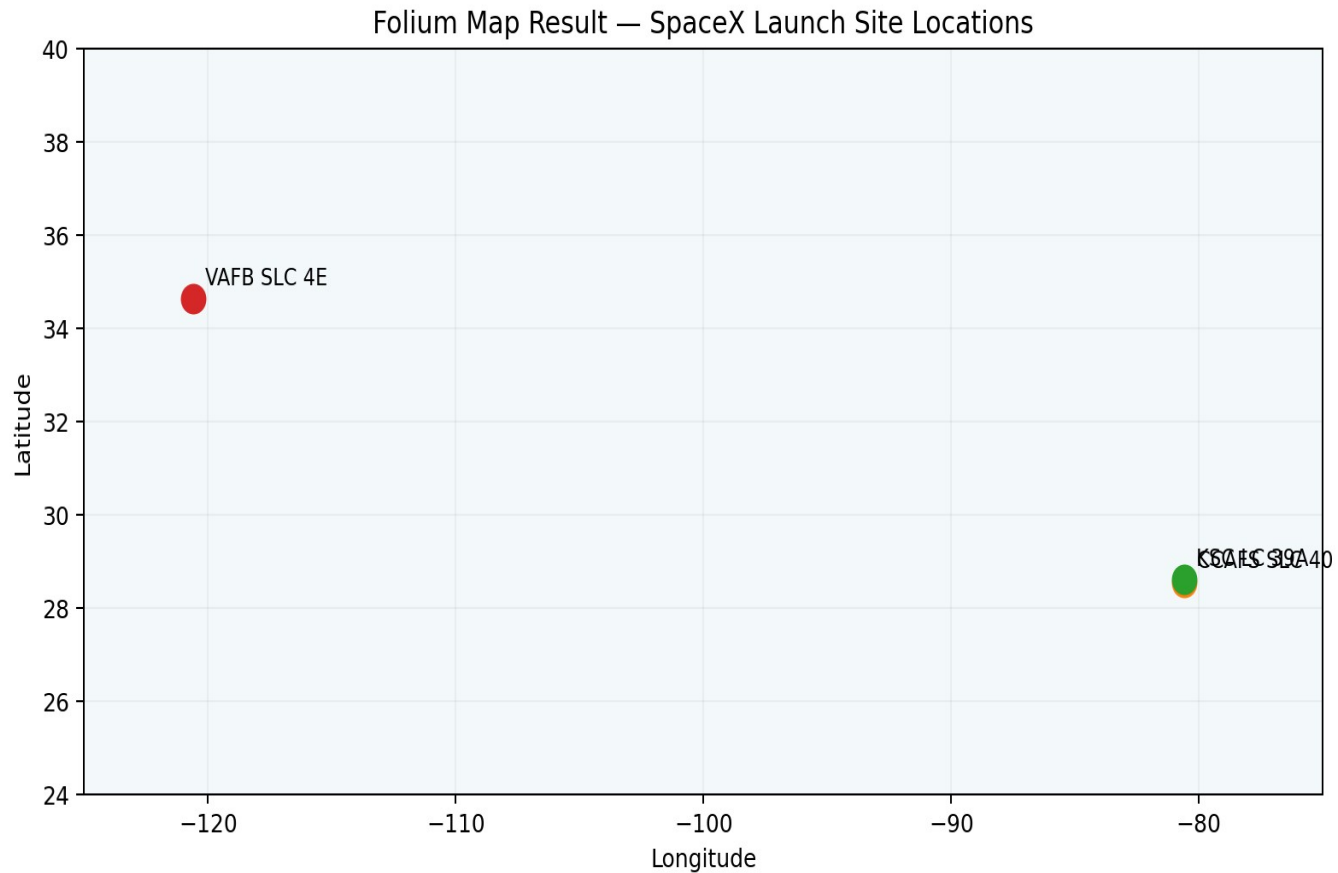
Notebook: [YOUR_REPOSITORY_URL](#)

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue header on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

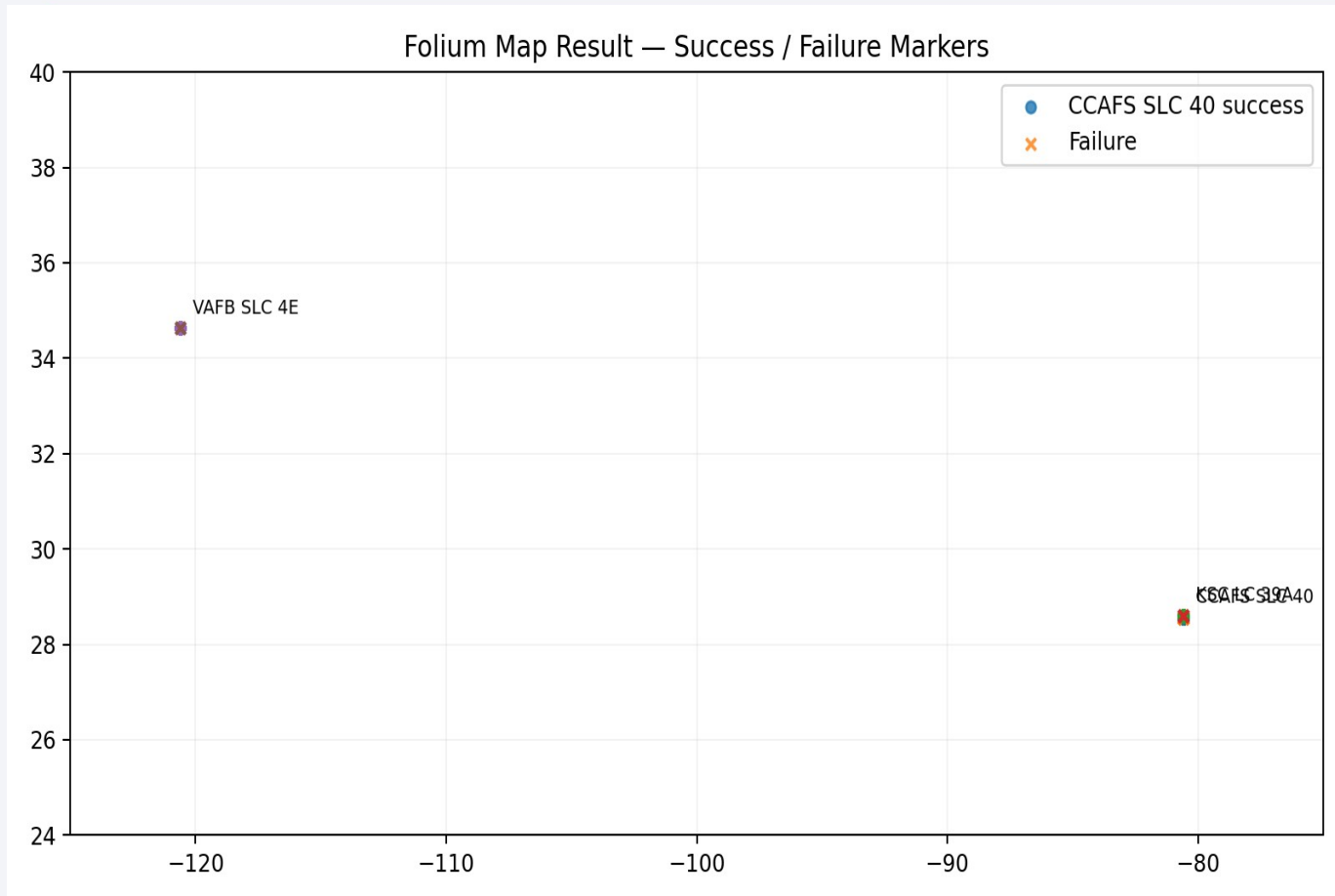
Folium Map – Launch Site Locations



Key findings

- Three orbital launch-site locations appear in Florida and California.
- All launch sites are coastal, consistent with over-ocean launch and stage-drop safety requirements.
- The map provides geographic context for the later outcome analysis.

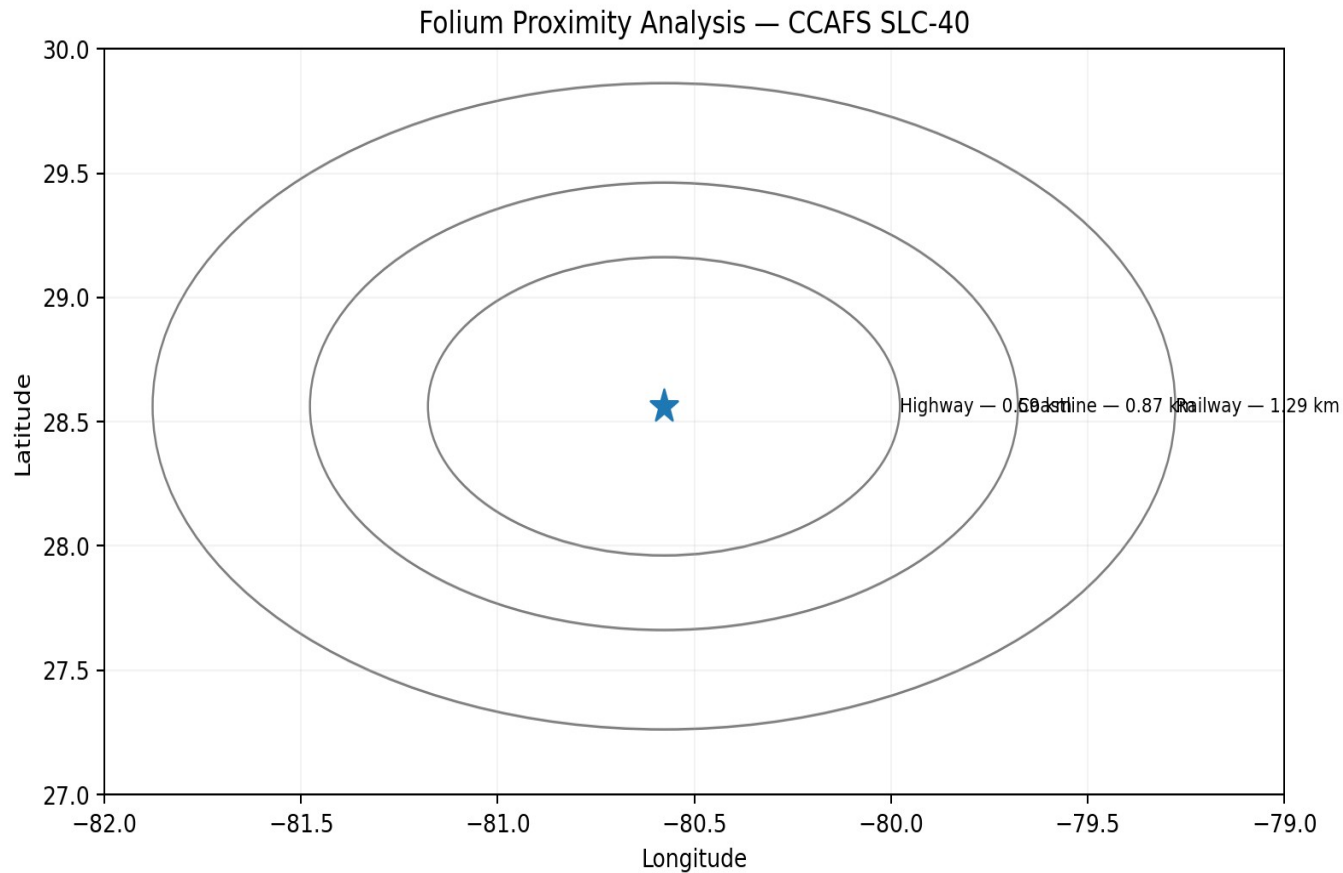
Folium Map – Success / Failure Outcomes



Key findings

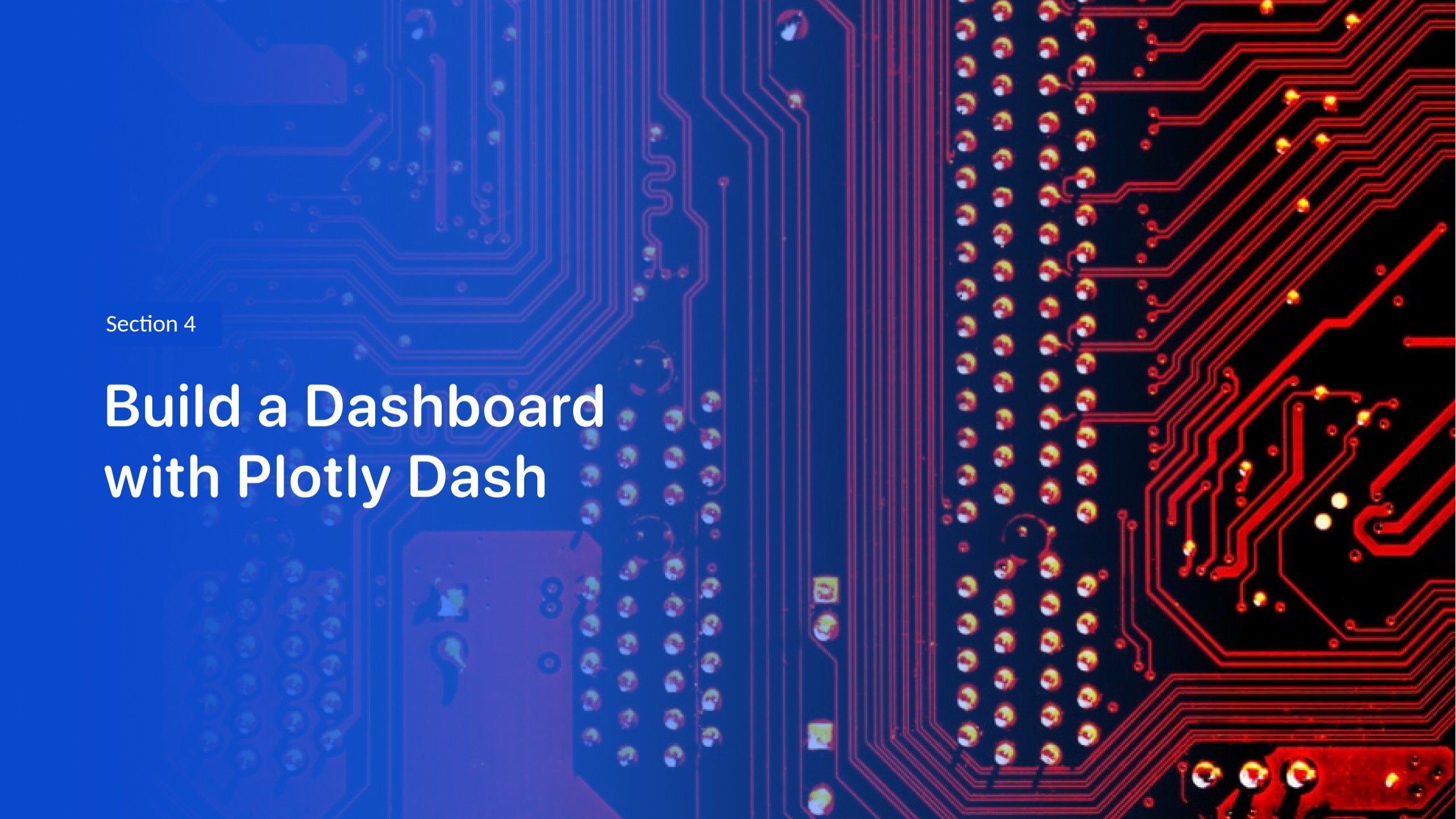
- Marker clustering reduces overlap when many launches share a site.
- Successful and failed outcomes are visually separated to expose site-level patterns.
- CCAFS SLC-40 has the largest launch volume; KSC LC-39A and VAFB SLC-4E show strong success shares.

Folium Proximity Analysis – CCAFS SLC-40



Key findings

- Reported proximity results: coastline $\approx$ 0.87 km; highway $\approx$ 0.59 km; railway $\approx$ 1.29 km; nearest city $\approx$ 51.74 km.
- The combination of coastal access and nearby transport infrastructure supports launch logistics.
- Distance annotations turn a location map into an operational planning view.

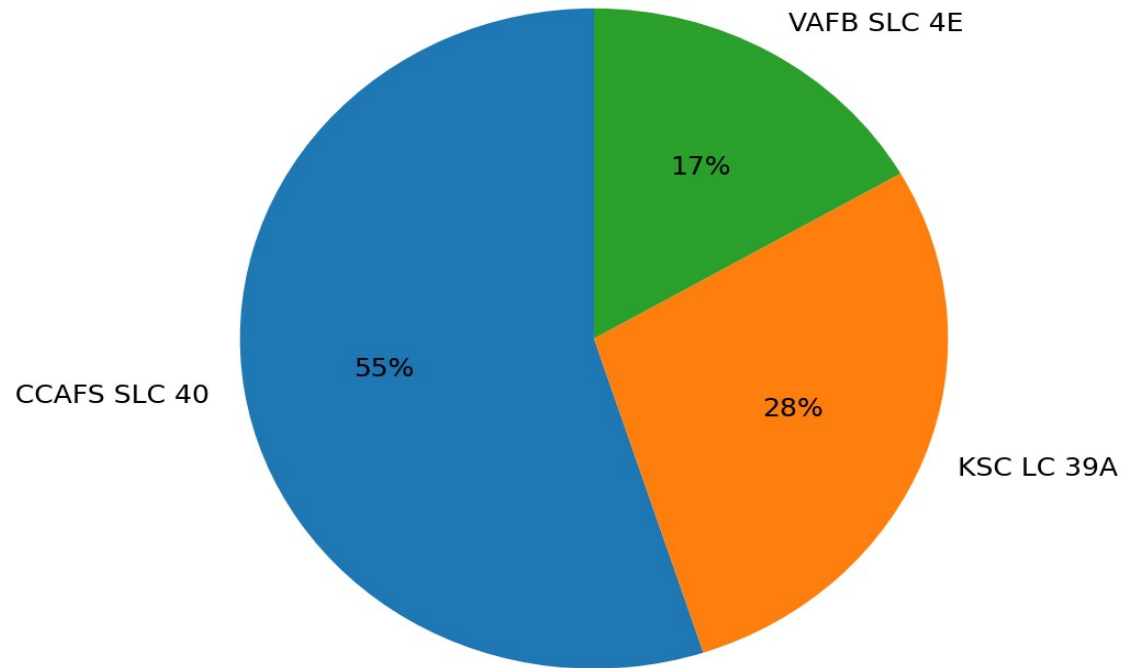


Section 4

Build a Dashboard with Plotly Dash

Plotly Dash – Launch Success Distribution

Launch Success Count by Site (Dashboard)

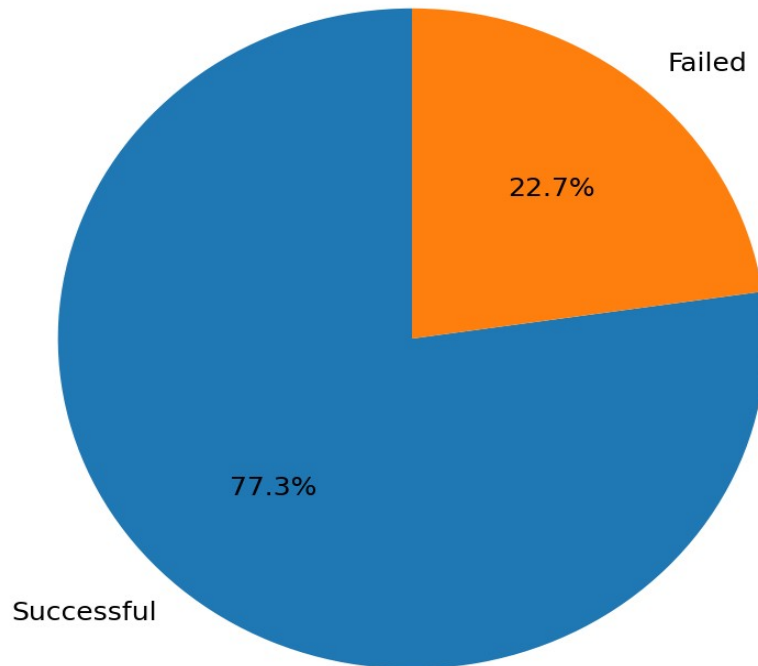


Dashboard insight

- The dashboard compares success counts across launch sites.
- CCAFS SLC-40 has the largest number of successful landings in the 90-record EDA dataset.
- The site selector allows the same view to be narrowed to one launch complex.

Plotly Dash – Highest Launch Success Ratio

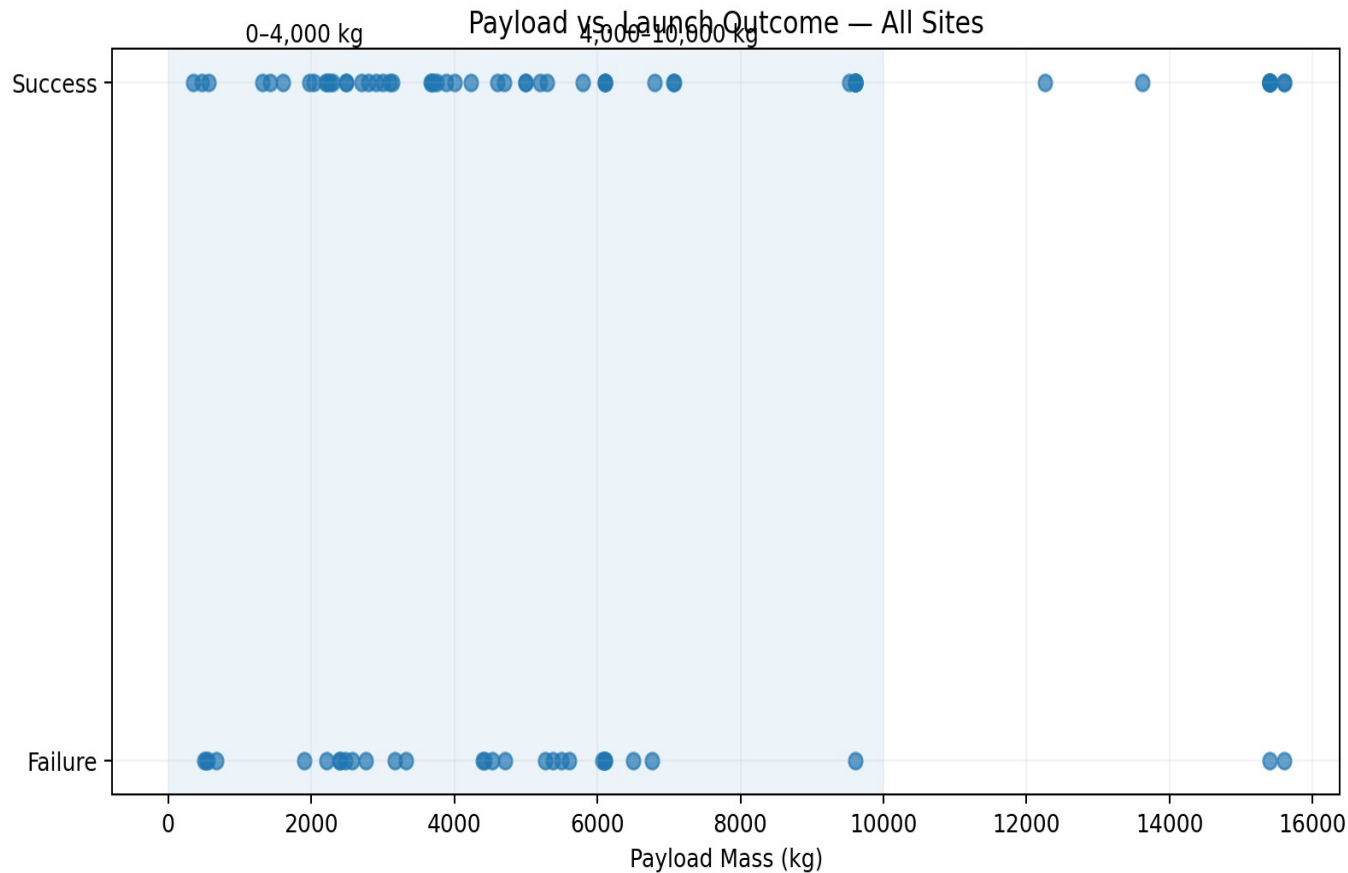
KSC LC-39A — Highest Launch Success Ratio (77.3%)



Dashboard insight

- KSC LC-39A has the highest site-level success ratio in the 90-record EDA dataset: $17/22 \approx 77.3\%$.
- This differs from launch-volume leadership: CCAFS SLC-40 has more total launches.
- The distinction between volume and reliability is operationally important.

Plotly Dash – Payload vs. Launch Outcome



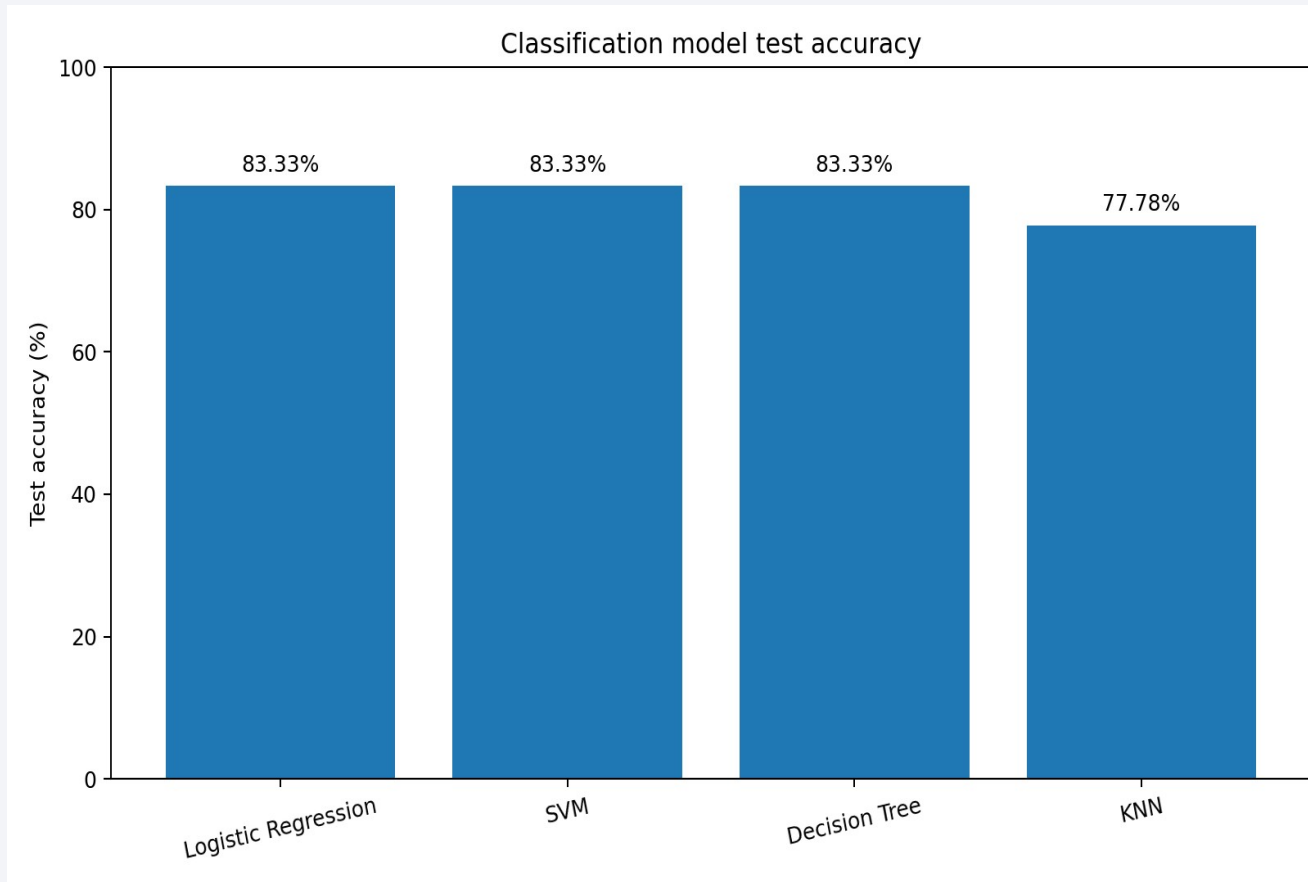
Dashboard insight

- A payload range control helps separate low- and high-payload mission profiles.
- The 0–4,000 kg region contains many successful launches; heavier missions are fewer and more heterogeneous.
- Booster generation and orbit interact with payload, so payload alone is not a sufficient predictor.

Section 5

Predictive Analysis (Classification)

Classification Accuracy



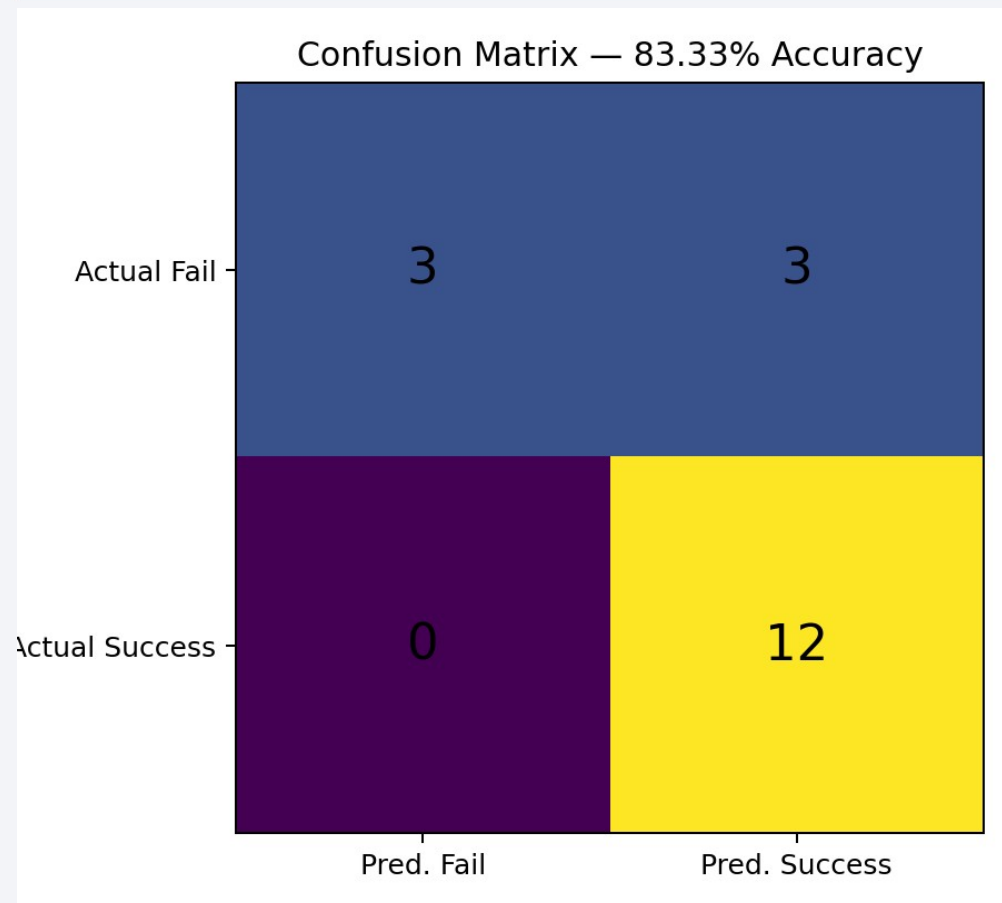
Result

Reproduced on the 90-row capstone extract with the standard 80/20 split (random_state=2).

- Logistic Regression: 83.33%
- SVM: 83.33%
- Decision Tree: 83.33%
- KNN: 77.78%

Decision Tree had the strongest 10-fold GridSearchCV score (~89.1%).

Confusion Matrix



- TN = 3: unsuccessful landings correctly identified.
- FP = 3: failures predicted as successful landings.
- FN = 0: no successful landings were missed.
- TP = 12: successful landings correctly identified.
- Accuracy = $(3 + 12) / 18 = 83.33\%$.

Model-selection insight: accuracy alone hides the cost of false positives; operational deployment should also consider precision, recall and the business cost of an incorrect landing prediction.

Conclusions

- Landing success improved substantially over time, especially after the early Falcon 9 flights.
- KSC LC-39A has the highest site-level landing success ratio in the analyzed 90-record dataset, while CCAFS SLC 40 has the highest launch volume.
- ES-L1, GEO, HEO and SSO show 100% success in this sample, while SO shows 0%; rare-orbit rates require cautious interpretation.
- Payload, orbit, launch site and flight history jointly provide useful predictive information.
- Four classification approaches achieved 83.33% test accuracy on the standard split; the small test set limits confidence in fine-grained model ranking.
- Interactive maps and dashboards make the analysis easier to explore and communicate than static tables alone.

Appendix

- Feature engineering: one-hot encoding of categorical variables.
- Landing label: Class = 1 successful first-stage landing; Class = 0 unsuccessful.
- SQL patterns: DISTINCT, LIKE, LIMIT, SUM, AVG, MIN, GROUP BY, COUNT, subqueries.
- Modeling: train/test split, scaling where required, GridSearchCV, confusion matrix.

Repository & external references

GitHub repository:
YOUR_REPOSITORY_URL

Primary analysis source:
IBM Applied Data Science Capstone — SpaceX Falcon 9 landing dataset and course workflow.

Before submission, replace YOUR_REPOSITORY_URL with your own completed GitHub repository.

Project References & Submission Checklist

- ✓ Executive summary, introduction and methodology included.
- ✓ Data collection, web scraping and wrangling documented.
- ✓ EDA visualization and SQL results included.
- ✓ Folium map and Plotly Dash results included.
- ✓ Classification accuracy and confusion matrix included.
- ✓ Conclusion and innovative insights included.

Thank you!

- ⚠ Replace YOUR_REPOSITORY_URL with the GitHub repository containing your completed notebooks / Python files before submission.

Final submission filename: Data Science Capstone Project Report.pdf